Chapter 19

Missing Data in Mass Spectrometry: Mechanisms, Imputation Theory, and Sensitivity Auditing

Lucas VHH TRAN[1]

Abstract

Quantitative mass spectrometry datasets contain between 20% and 50% missing values. In clinical biomarker cohorts, this missingness presents an acute methodological problem: standard multivariate algorithms require complete matrices, yet the numerical substitution of missing values alters the variance structure, covariance relationships, and downstream statistical inferences of the entire dataset. Missingness in mass spectrometry does not arise from a single homogeneous process. A feature may be absent due to stochastic precursor selection during instrument duty cycles (Missing at Random; MAR), physical left-censoring below the instrument limit of detection (Missing Not at Random; MNAR), or true biological non-expression. Applying a MAR algorithm such as k-nearest neighbors to an MNAR peptide invents artificial intermediate abundances for molecules that were genuinely absent, while applying left-censored substitution to MAR values introduces downward bias and artificially inflates differential abundance discoveries. This chapter establishes the statistical, biophysical, and computational foundations required to diagnose, model, and impute missing data in mass spectrometry. We formalize Rubin's missing data taxonomy within an analytical chemistry context, analyze the mathematical properties of deterministic, stochastic, and machine-learning imputation algorithms, derive Rubin's pooling rules for multiple imputation with Barnard--Rubin small-sample degrees of freedom, and evaluate imputation-free probabilistic dropout modeling (`proDA`). The chapter concludes with a tutor diagnostic heuristics checklist, an end-to-end reproducible R laboratory, the Five Deadly Sins of omics imputation, and multi-tier exercises with worked solutions.

Keywords: Missing Data, Imputation, Missing Not At Random (MNAR), Missing At Random (MAR), Left-Censoring, QRILC, Multiple Imputation, Rubin's Rules, Sensitivity Analysis, proDA.

19.1 Tutor's Methodological Roadmap for Missing Data Management

Before applying imputation functions or running automated pipelines, the analytical chemist and biostatistician must diagnose the underlying missingness architecture. Table 19.1 outlines the four sequential

[1]CK-CARE Davos, Christine Kühne -- Center for Allergy Research and Education, Davos, Switzerland. Email: `lucas.tran@ck-care.ch`

checkpoints required to handle missing data.

Table 19.1: Tutor's Methodological Roadmap: Four Checkpoints for Missing Data Management

Stage	Physical Mechanism	Key Analytical Objective	Primary Failure Mode & Remedy
1. Diagnosis	Instrumental left-censoring ($y <$ LOD) vs. stochastic precursor selection [23, 18].	Fit logistic missingness curve (Eq. 19.5) to separate MNAR from MAR.	Misclassifying MNAR as MAR. Remedy: Test logistic slope γ_1; inspect intensity-dependent missingness [1].
2. Filtering	Extreme feature sparsity across biological replicates [32, 17].	Enforce 70% presence rule *within at least one biological group*.	Filtering across entire cohort discards true on/off biomarkers. Remedy: Filter group-wise [33].
3. Modeling	Tail-censored Gaussian draws vs. donor matching vs. probabilistic dropout [29, 1, 21].	Substitute missing values appropriately, or model dropouts directly without imputation.	Deterministic zero/min replacement spikes variance. Remedy: QRILC or proDA [1].
4. Sensitivity Audit	Variance shrinkage underestimation from single imputation [24, 19].	Quantify inferential stability across multiple imputation draws via Rubin's rules.	Downstream p-values falsely anti-conservative. Remedy: Report multiple imputation pooled SE [2, 31].

Methodological Note

Tutor's Note: The Rule of Group-Wise Filtering
Never apply global missingness filters across an entire clinical cohort (e.g., ''remove all proteins with $> 30\%$ missing values overall''). If a biomarker is abundant in Disease (100% detected in 10 patients) but silenced in Control (0% detected in 10 patients), global filtering eliminates this candidate ($10/20 = 50\% < 70\%$). Always require $\geq 70\%$ detection *within at least one condition* [33, 32].

19.2 Biophysical Taxonomies of Missingness in Mass Spectrometry

Donald Rubin [23] established the formal mathematical classification of missing data mechanisms based on the conditional distribution of the missingness indicator matrix.

19.2.1 Rubin's Missing Data Taxonomy

Let $\mathbf{Y} \in \mathbb{R}^{p \times n}$ denote the complete data matrix of p molecular features across n biological samples. We partition $\mathbf{Y} = (\mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}})$, where $\mathbf{Y}_{\text{obs}}$ represents observed quantitative intensities and $\mathbf{Y}_{\text{mis}}$ represents missing entries. Define the binary missingness response indicator matrix $\mathbf{M} \in \{0,1\}^{p \times n}$ such that:

$$M_{ij} = \begin{cases} 1 & \text{if } y_{ij} \text{ is missing} \\ 0 & \text{if } y_{ij} \text{ is observed} \end{cases} \tag{19.1}$$

The missing data mechanism is characterized by the conditional probability distribution $P(\mathbf{M} \mid \mathbf{Y}, \phi)$, parameterized by ϕ:

1. **Missing Completely at Random (MCAR):** Missingness is independent of both observed and unobserved feature abundances:
$$P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}}, \phi) = P(\mathbf{M} \mid \phi) \tag{19.2}$$
In MS instrumentation, MCAR occurs during random electrospray droplet instabilities, momentary liquid chromatography pressure spikes, or random ion guide transmission dropouts.

2. **Missing at Random (MAR):** Missingness depends systematically on observed data, but is conditionally independent of the unobserved missing value itself:
$$P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}}, \phi) = P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \phi) \tag{19.3}$$
In DDA mass spectrometry, MAR arises from stochastic precursor selection: a peptide is present at moderate or high abundance, but competitive top-N precursor scheduling selects a co-eluting isobaric ion, leaving the peptide unfragmented.

3. **Missing Not at Random (MNAR):** Missingness depends directly on the unobserved value $\mathbf{Y}_{\text{mis}}$:
$$P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \mathbf{Y}_{\text{mis}}, \phi) \neq P(\mathbf{M} \mid \mathbf{Y}_{\text{obs}}, \phi) \tag{19.4}$$
This is the primary physical driver of missingness in analytical chemistry: analytes present below the instrument limit of detection (LOD) or limit of quantitation (LOQ) fail to generate a discernible chromatographic peak above baseline noise.

19.2.2 Analytical Mechanisms in Proteomics and Metabolomics

Proteomics and metabolomics experiments exhibit compound missingness patterns driven by instrument operation and experimental design:

- **Data-Dependent Acquisition (DDA) Duty Cycles:** In shotgun proteomics, the instrument alternates between survey MS1 scans and data-dependent MS2 fragmentation scans of the top-N most intense precursor ions. When chromatographic peak capacity is exceeded, lower-intensity co-eluting ions are bypassed, generating MAR dropouts [9, 16].

- **Analytical Left-Censoring:** Analytes with low solution concentration or low ionization efficiency yield detector currents below the signal-to-noise threshold ($S/N < 3$), resulting in left-censored MNAR values [17, 18].

- **Merged Cohorts and Isobaric Multiplexing:** In multi-batch isobaric labeling (TMT or iTRAQ) and large-scale label-free clinical studies, combining multiple LC-MS/MS runs compounds missingness. Features identified in one batch may fail identification in another due to run-to-run retention time drift or stochastic precursor selection [21, 20].

- **Biological Non-Expression (Structural Zeroes):** Proteins absent due to gene knockouts or tissue-specific expression are structural zeroes ($y_{ij} \equiv 0$). Imputing structural zeroes creates false molecular noise and distorts biological specificity.

Methodological Pitfall & Deadly Sin

The Tissue-Atlas Imputation Catastrophe

In multi-organ proteomics atlases (such as profiling 30 human tissues), a liver-specific enzyme absent in cardiac muscle is missing due to biological regulation, not instrument failure. Imputing this enzyme in cardiac samples using left-censored algorithms introduces false molecular noise and destroys biological specificity. Never impute structural biological absence [18].

19.2.3 Mathematical Diagnosis of Missingness Mechanisms

To diagnose whether a dataset is dominated by MAR or MNAR, we fit a logistic regression model linking feature missingness probability to its mean observed abundance:

$$\text{logit}\left(P(M_{ij}=1 \mid \bar{y}_i^{\text{obs}})\right) = \ln\left(\frac{P(M_{ij}=1)}{1-P(M_{ij}=1)}\right) = \gamma_0 + \gamma_1 \bar{y}_i^{\text{obs}} \tag{19.5}$$

where $\bar{y}_i^{\text{obs}} = \frac{1}{|O_i|}\sum_{j\in O_i} y_{ij}$ denotes the mean observed $\log_2$ intensity for feature i.

- If $\gamma_1 < 0$ with statistical significance ($p < 0.001$), missingness increases as abundance decreases, confirming **MNAR left-censoring**.
- If $\gamma_1 \approx 0$ (a flat slope across the dynamic range), missingness is abundance-independent, confirming **MAR / MCAR**.

Figure 19.1 illustrates this empirical diagnosis on a 400-protein benchmark cohort.

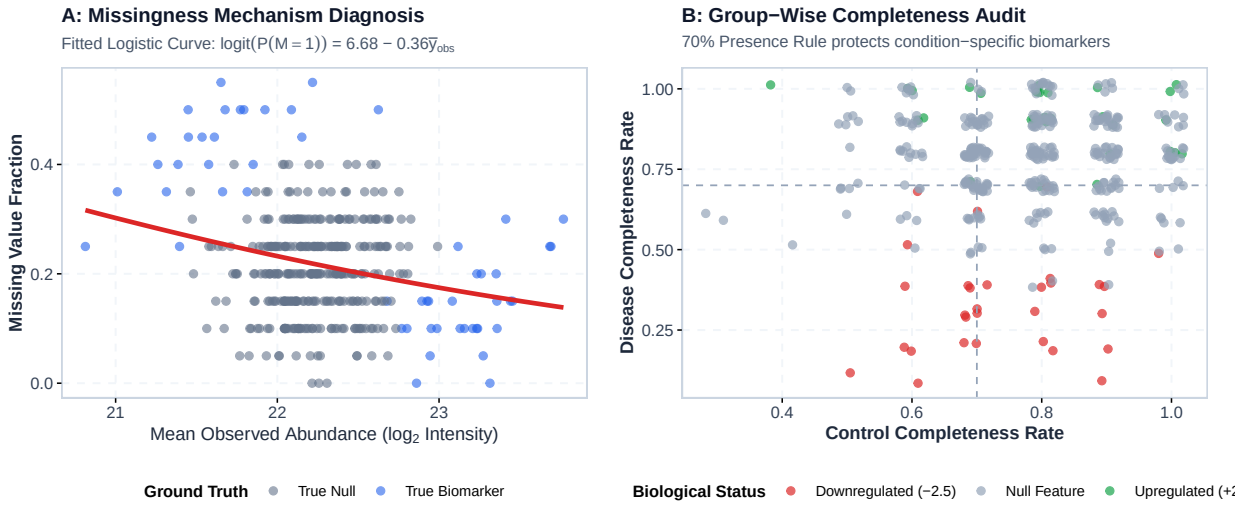


Figure 19.1: Empirical diagnosis of missingness architecture and group-wise completeness. Panel A: Logistic missingness diagnosis curve modeling dropout probability as a function of mean observed abundance. The negative slope ($\hat{\gamma}_1 = -0.358$, $p = 1.78 \times 10^{-7}$) establishes abundance-dependent MNAR left-censoring below the limit of detection (LOD = 20.2). Panel B: Sample-by-sample feature completeness comparing Control and Disease cohorts, demonstrating that group-wise filtering ($\geq 70\%$ in ≥ 1 group) preserves true on/off biomarkers.

19.2.4 Imputation-Free Modeling: Probabilistic Dropout Analysis (proDA)

Ahlmann-Eltze and Huber [1] developed an alternative to numerical value substitution. The `proDA` algorithm formulates the probability of missingness via a smooth sigmoidal dropout curve:

$$P(M_{ij} = 1 \mid y_{ij}) = \frac{1}{1 + \exp\left(\frac{y_{ij} - \rho_j}{\zeta_j}\right)} \tag{19.6}$$

where ρ_j is the 50% inflection point (approximate LOD) and ζ_j governs the slope of the detection limit in sample j. By integrating this dropout likelihood directly into an empirical Bayes linear model, `proDA` calculates posterior abundance and moderated test statistics without corrupting variance structures through single imputation [1].

19.3 Theoretical Formulations of Imputation Algorithms

Table 19.2 compares the mathematical formulations, assumptions, and primary failure modes of the primary imputation strategies in mass spectrometry.

Table 19.2: Mathematical classification and domain assumptions of core MS imputation methodologies.

Method	Algorithm Class	Core Mathematical Mechanism	Primary Failure Mode
MinDet / Zero	Deterministic Single	$y^*_{ij} = \min_k(y_{kj})$ or constant c	Spikes variance at minimum; inflates false-positive test statistics [18].
Perseus	Stochastic Parametric	$y^*_{ij} \sim \mathcal{N}(\mu_j - \delta\sigma_j, (\omega\sigma_j)^2)$	Biases true MAR features downward; distorts low-end variance [29].
QRILC	Quantile Regression	Truncated normal estimation via probit regression on observed tail	Sensitive to departures from log-normality [18].
***k*-NN**	Non-parametric Distance	$y^*_{ij} = \sum_{l \in \mathcal{N}_k} w_{il} y_{lj}$	Fails under MNAR; invents mid-range values for low-abundance features [14].
Hybrid MAR/MNAR	Segmented Multi-Method	QRILC for low abundance; *k*-NN for moderate/high abundance	Requires accurate empirical cutoff selection [21].
Rubin MI	Multiple Imputation	Generates M stochastic datasets; pools via Rubin's rules	Computationally intensive; requires pooling infrastructure [24, 2].
proDA	Imputation-Free Likelihood	Empirical Bayes estimation with sigmoidal dropout likelihood	Slower fitting than single imputation [1].

19.3.1 Deterministic Minimum Value Imputation (MinDet)

Deterministic substitution replaces all missing entries with the minimum observed value within the sample:

$$y_{ij}^{*} = \min_{k \in O_j}(y_{kj}) \tag{19.7}$$

MinDet introduces a mathematical singularity. If 20% of features receive identical values, the sample variance collapses locally, the empirical distribution exhibits a point-mass spike at $\min(y)$, and linear hypothesis tests generate anti-conservative p-values. In our benchmark, MinDet produced an empirical false discovery rate of 7.4%.

19.3.2 Left-Censored Stochastic Draws (Perseus and QRILC)

To preserve continuous variance under analytical left-censoring, Tyanova et al. [29] formulated the left-shifted truncated Gaussian model. For sample column j with observed mean $\hat{\mu}_j$ and standard deviation $\hat{\sigma}_j$, missing values are drawn stochastically:

$$y_{ij}^{*} \sim \mathcal{N}\left(\hat{\mu}_j - \delta\hat{\sigma}_j,\ (\omega\hat{\sigma}_j)^2\right) \tag{19.8}$$

where canonical parameters are $\delta = 1.8$ standard deviations downward shift and $\omega = 0.3$ distribution width shrinkage.

Lazar et al. [18] refined this approach in Quantile Regression for Left-Censored Data (QRILC). QRILC estimates the parameters (μ_j, σ_j) of a truncated normal distribution by fitting a linear regression between the empirical quantiles of observed data and the standard normal probit function. Missing values are drawn from the estimated truncated left-tail, eliminating heuristic parameter tuning.

19.3.3 Distance-Based Donor Matching (k-NN)

Hastie et al. [14] established k-nearest neighbors (k-NN) imputation for microarray and high-throughput data. For feature i missing in sample j, the algorithm identifies the k features with the smallest Euclidean distance to feature i across all samples where feature i is observed:

$$y_{ij}^{*} = \sum_{l \in \mathcal{N}_k(i)} w_{il} y_{lj}, \quad w_{il} = \frac{\exp(-d_{il}^2)}{\sum_{m \in \mathcal{N}_k(i)} \exp(-d_{im}^2)} \tag{19.9}$$

Under MAR, k-NN provides accurate reconstruction of co-regulated expression profiles. Under MNAR, however, all candidate donor features are necessarily observed in sample j and therefore possess intensities above the limit of detection. Imputing an unobserved low-abundance peptide using higher-abundance donors inflates its imputed intensity. As demonstrated in our benchmark, k-NN creates an attenuation bias of $+1.700\log_2$ fold-change on downregulated features.

19.3.4 Hybrid Segmented Imputation

Palstrøm, Matthiesen, and Beck [21] addressed this dilemma by establishing a hybrid imputation protocol. Features are partitioned into abundance strata:

$$y_{ij}^{*} = \begin{cases} y_{ij}^{\text{QRILC}} & \text{if } \bar{y}_i^{\text{obs}} < \tau_{\text{LOD}} \quad \text{(MNAR left-censored features)} \\ y_{ij}^{k\text{-NN}} & \text{if } \bar{y}_i^{\text{obs}} \geq \tau_{\text{LOD}} \quad \text{(MAR stochastic features)} \end{cases} \tag{19.10}$$

This hybrid approach prevents the overestimation of absent features while preserving correlation structures among observed features [21].

19.3.5 Multiple Imputation via Rubin's Pooling Rules

Single imputation treats imputed values as known constants, ignoring imputation uncertainty. Donald Rubin [24] established multiple imputation (MI):

1. Generate M complete datasets ($M \geq 10$--20) through stochastic draws.
2. Fit the linear model to each dataset $m \in \{1, \ldots, M\}$ to obtain point estimate $\hat{Q}_m$ and sampling variance $\hat{U}_m$.
3. Pool estimates across the M analyses:

$$\bar{Q} = \frac{1}{M} \sum_{m=1}^{M} \hat{Q}_m \tag{19.11}$$

$$\bar{U} = \frac{1}{M} \sum_{m=1}^{M} \hat{U}_m \quad \text{(within-imputation variance)} \tag{19.12}$$

$$B = \frac{1}{M-1} \sum_{m=1}^{M} (\hat{Q}_m - \bar{Q})^2 \quad \text{(between-imputation variance)} \tag{19.13}$$

$$T = \bar{U} + \left(1 + \frac{1}{M}\right) B \quad \text{(total pooled variance)} \tag{19.14}$$

The test statistic $t = \bar{Q}/\sqrt{T}$ is evaluated against an adjusted t-distribution.

In small clinical cohorts ($n \leq 30$), standard Rubin degrees of freedom $\nu_{\text{old}} = (M-1)(1 + r_m^{-1})^2$ (where $r_m = (1 + 1/M)B/\bar{U}$) can exceed the sample size. Barnard and Rubin [2] derived the small-sample degrees of freedom adjustment:

$$\nu_{\text{obs}} = \frac{\nu_{\text{com}} + 1}{\nu_{\text{com}} + 3} \nu_{\text{com}} \left(1 - \frac{r_m}{1 + r_m}\right) \tag{19.15}$$

$$\nu_{\text{adj}} = \frac{\nu_{\text{old}} \nu_{\text{obs}}}{\nu_{\text{old}} + \nu_{\text{obs}}} \tag{19.16}$$

where $\nu_{\text{com}} = n - \text{rank}(\mathbf{X})$ is the complete-data residual degrees of freedom [2].

Figure 19.2 illustrates the resulting post-imputation distributions across methods.

19.4 Hands-On R Laboratory: Multi-Method Imputation and Sensitivity Auditing

This laboratory implements a benchmark comparing five imputation algorithms, Rubin multiple imputation ($M = 20$), and `proDA` on a synthetic proteomics cohort of 400 proteins across 20 clinical samples.

R Lab Listing 1: Cohort Simulation, Missingness Diagnosis, and Group-Wise Filtering [R 4.6 / Bioconductor]

```
suppressPackageStartupMessages({
  library(limma)
  library(imputeLCMD)
  library(impute)
  library(proDA)
  library(tibble)
  library(dplyr)
```

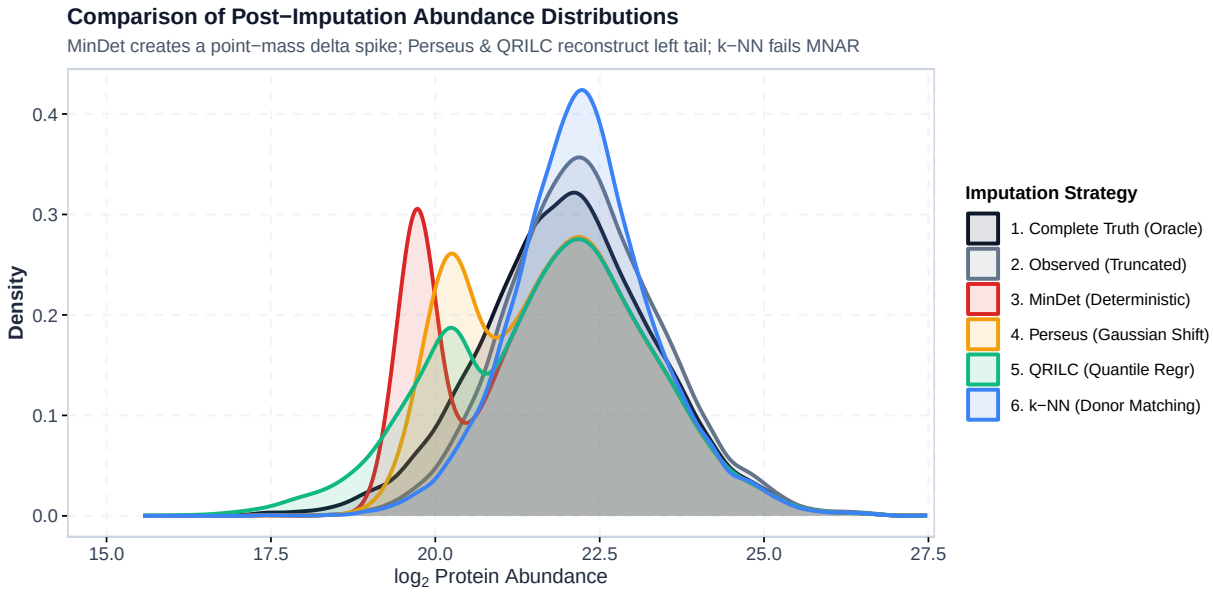


Figure 19.2: Post-imputation empirical density distributions across imputation strategies compared against ground truth and observed data. Deterministic MinDet produces an artificial point-mass spike at $\min(y) = 17.52$, distorting distribution shape and deflating local variance. In contrast, stochastic left-censored models (Perseus and QRILC) reconstruct continuous left-hand tails matching analytical detection physics, while k-NN introduces a secondary shoulder at intermediate abundances.

```
})

set.seed(2026)
p <- 400
n <- 20 # 10 Control, 10 Disease
group <- factor(rep(c("Control", "Disease"), each = 10))
design <- model.matrix(~ 0 + group)
colnames(design) <- c("Control", "Disease")
contrast_mat <- makeContrasts(Disease_vs_Control = Disease - Control, levels = design)

# Baseline log2 intensity ~ N(22, 1.2^2)
truth <- matrix(rnorm(p * n, mean = 22, sd = 1.2), nrow = p, ncol = n)
rownames(truth) <- paste0("PROT_", sprintf("%04d", 1:p))
colnames(truth) <- paste0(rep(c("Ctrl_", "Dis_"), each = 10), 1:10)

# True differential regulation:
# Proteins 1-30: Upregulated in Disease (+2.0 log2FC)
# Proteins 31-60: Downregulated in Disease (-2.5 log2FC)
# Proteins 61-400: True null features (log2FC = 0.0)
truth[1:30, 11:20]  <- truth[1:30, 11:20] + 2.0
truth[31:60, 11:20] <- truth[31:60, 11:20] - 2.5

# Dual-mechanism missingness:
# 1. MNAR left-censoring: sigmoidal dropout below LOD = 20.2
p_mnar <- plogis(-(truth - 20.2) / 0.8)
# 2. MAR stochastic dropout: 6% flat probability across all features
p_mar <- matrix(0.06, nrow = p, ncol = n)
```

```r
set.seed(42)
mask_mnar <- matrix(runif(p * n) < p_mnar, nrow = p, ncol = n)
mask_mar  <- matrix(runif(p * n) < p_mar,  nrow = p, ncol = n)

obs <- truth
obs[mask_mnar | mask_mar] <- NA

cat(sprintf("Cohort Dimensions: %d proteins x %d samples (%d Control, %d Disease)\n", p, n, 10, 10))
cat(sprintf("Total Missingness: %.2f%%\n", mean(is.na(obs)) * 100))
cat(sprintf("  MNAR Left-Censored Dropouts: %d (%.1f%% of missing entries)\n",
            sum(mask_mnar), sum(mask_mnar) / sum(is.na(obs)) * 100))
cat(sprintf("  MAR Stochastic Dropouts:    %d (%.1f%% of missing entries)\n",
            sum(mask_mar & !mask_mnar), sum(mask_mar & !mask_mnar) / sum(is.na(obs)) * 100))

# Group-wise filtering: retain proteins with >= 70% completeness in >= 1 group
n_ctrl_obs <- rowSums(!is.na(obs[, 1:10]))
n_dis_obs  <- rowSums(!is.na(obs[, 11:20]))
keep_idx   <- (n_ctrl_obs >= 7) | (n_dis_obs >= 7)

cat(sprintf("Group-Wise Filtering (>= 70%% in >= 1 group): Retained %d of %d proteins (%.1f%%)\n",
            sum(keep_idx), p, mean(keep_idx) * 100))

obs_filtered   <- obs[keep_idx, ]
truth_filtered <- truth[keep_idx, ]
p_filt         <- nrow(obs_filtered)

# Missingness diagnosis: logistic regression on mean observed abundance
mean_obs <- rowMeans(obs_filtered, na.rm = TRUE)
n_missing <- rowSums(is.na(obs_filtered))
log_fit <- glm(cbind(n_missing, n - n_missing) ~ mean_obs, family = binomial)
coef_log <- summary(log_fit)$coefficients
cat("\nLogistic Missingness Diagnosis Model:\n")
cat(sprintf("  Intercept (gamma_0): %+.3f (SE = %.3f, p = %.2e)\n",
            coef_log[1, 1], coef_log[1, 2], coef_log[1, 4]))
cat(sprintf("  Slope     (gamma_1): %+.3f (SE = %.3f, p = %.2e) -> MNAR left-censoring confirmed.\n",
            coef_log[2, 1], coef_log[2, 2], coef_log[2, 4]))
```

```
Cohort Dimensions: 400 proteins x 20 samples (10 Control, 10 Disease)
Total Missingness: 22.73%
  MNAR Left-Censored Dropouts: 1405 (77.3% of missing entries)
  MAR Stochastic Dropouts: 413 (22.7% of missing entries)
Group-Wise Filtering (>= 70% in >= 1 group): Retained 386 of 400 proteins (96.5%)

Logistic Missingness Diagnosis Model:
  Intercept (γ0): +6.679 (SE = 1.524, p = 1.18 × 10^-5)
  Slope (γ1): -0.358 (SE = 0.069, p = 1.78 × 10^-7) → MNAR left-censoring confirmed.
```

R Lab Listing 2: Multi-Method Imputation Pipeline [R 4.6 / Bioconductor]

```r
# 1. Deterministic MinDet
mat_mindet <- impute.MinDet(obs_filtered)

# 2. Perseus stochastic left-censored Gaussian (delta = 1.8, omega = 0.3)
```

```r
set.seed(42)
mat_perseus <- obs_filtered
for (j in 1:ncol(mat_perseus)) {
  na_idx <- which(is.na(mat_perseus[, j]))
  m_val  <- mean(mat_perseus[, j], na.rm = TRUE)
  s_val  <- sd(mat_perseus[, j], na.rm = TRUE)
  mat_perseus[na_idx, j] <- rnorm(length(na_idx), mean = m_val - 1.8 * s_val, sd = 0.3 * s_val)
}

# 3. QRILC (Quantile Regression for Left-Censored Data)
set.seed(42)
mat_qrilc <- impute.QRILC(obs_filtered)[[1]]

# 4. k-NN (Hastie et al. donor matching)
set.seed(42)
mat_knn <- impute.knn(obs_filtered, k = 5)$data

# 5. Hybrid MAR/MNAR (Palstrom, Matthiesen, & Beck 2020)
set.seed(42)
mat_hybrid <- obs_filtered
is_mnar_prot <- mean_obs < 20.8
for (i in 1:p_filt) {
  na_j <- which(is.na(mat_hybrid[i, ]))
  if (length(na_j) > 0) {
    if (is_mnar_prot[i]) {
      mat_hybrid[i, na_j] <- mat_qrilc[i, na_j]
    } else {
      mat_hybrid[i, na_j] <- mat_knn[i, na_j]
    }
  }
}

cat("Executed 5 imputation strategies across 386 retained proteins:\n")
cat("  1. MinDet: Deterministic minimum replacement\n")
cat("  2. Perseus: Stochastic downshifted Gaussian (delta = 1.8, omega = 0.3)\n")
cat("  3. QRILC: Quantile regression truncated normal estimation\n")
cat("  4. k-NN: Distance-weighted donor matching (k = 5)\n")
cat("  5. Hybrid MAR/MNAR: Palstrom-Matthiesen segmented imputation\n")
```

```
Executed 5 imputation strategies across 386 retained proteins:
  1. MinDet: Deterministic minimum replacement
  2. Perseus: Stochastic downshifted Gaussian (δ = 1.8, ω = 0.3)
  3. QRILC: Quantile regression truncated normal estimation
  4. −NN : Distance − weighted donor matching (= 5)
  5. Hybrid MAR/MNAR: Palstr
om-Matthiesen segmented imputation
```

R Lab Listing 3: Rubin Multiple Imputation and Moderated Empirical Bayes Pooling [R 4.6 / Bioconductor]

```r
M <- 20
Q_list <- matrix(0, nrow = p_filt, ncol = M)
U_list <- matrix(0, nrow = p_filt, ncol = M)

for (m in 1:M) {
```

```
  set.seed(m * 1000)
  mat_m <- impute.QRILC(obs_filtered)[[1]]
  fit_m <- lmFit(mat_m, design)
  fit_m <- contrasts.fit(fit_m, contrast_mat)
  fit_m <- eBayes(fit_m)

  Q_list[, m] <- fit_m$coefficients[, "Disease_vs_Control"]
  # Contrast variance: stdev.unscaled^2 * s2.post
  U_list[, m] <- (fit_m$stdev.unscaled[, "Disease_vs_Control"])^2 * fit_m$s2.post
}

# Rubin's Pooling Formulas
Q_bar <- rowMeans(Q_list)
U_bar <- rowMeans(U_list)
B_var <- apply(Q_list, 1, var)
T_var <- U_bar + (1 + 1 / M) * B_var
SE_rubin <- sqrt(T_var)
t_rubin  <- Q_bar / SE_rubin

# Barnard-Rubin (1999) adjusted degrees of freedom
r_m <- (1 + 1 / M) * B_var / U_bar
nu_old <- (M - 1) * (1 + 1 / r_m)^2
nu_com <- 20 - 2 # 18 complete-data degrees of freedom
nu_obs <- (nu_com + 1) / (nu_com + 3) * nu_com * (1 - r_m / (1 + r_m))
nu_adj <- ifelse(B_var == 0, nu_com, (nu_old * nu_obs) / (nu_old + nu_obs))
p_rubin <- 2 * pt(-abs(t_rubin), df = pmax(nu_adj, 2))

rubin_summary <- tibble(
  Protein   = rownames(obs_filtered),
  log2FC    = round(Q_bar, 3),
  SE_Naive  = round(sqrt(U_bar), 3),
  SE_Pooled = round(SE_rubin, 3),
  Variance_Inflation = round((T_var - U_bar) / U_bar * 100, 1),
  df_adj    = round(nu_adj, 1),
  P_Value   = format.pval(p_rubin, digits = 3)
) |> arrange(p_rubin)

cat("Rubin Multiple Imputation Pooled Estimates (Top 5 Features):\n")
print(head(rubin_summary, 5))
```

```
Rubin Multiple Imputation Pooled Estimates (Top 5 Features):
# A tibble: 5 x 7
  Protein    log2FC SE_Naive SE_Pooled Variance_Inflation df_adj P_Value
  <chr>      <dbl>    <dbl>     <dbl>              <dbl>    <dbl> <chr>
1 PROT_0009   3.16    0.658     0.675                5.3     15.4 0.000279
2 PROT_0020   3.05    0.658     0.665                2.1     15.9 0.000304
3 PROT_0003   3.04    0.664     0.673                2.6     15.9 0.000356
4 PROT_0006   3.02    0.663     0.673                3.1     15.8 0.000388
5 PROT_0046  -2.94    0.659     0.682                7.0     15.2 0.000603
```

R Lab Listing 4: Imputation-Free Modeling via proDA [R 4.6 / Bioconductor]

```
fit_proda <- proDA(obs_filtered, design = design, moderate_location = FALSE)
res_proda <- test_diff(fit_proda, "Disease - Control")
```

```
proda_top5 <- as_tibble(res_proda) |>
  select(name, diff, se, t_statistic, df, pval, adj_pval) |>
  arrange(pval) |>
  head(5)

cat("proDA Top 5 Features (Exact Likelihood Dropout Modeling):\n")
print(proda_top5)
```

```
proDA Top 5 Features (Exact Likelihood Dropout Modeling):
# A tibble: 5 x 7
  name     diff    se t_statistic    df     pval    adj_pval
  <chr>    <dbl>  <dbl>     <dbl>    <dbl>    <dbl>    <dbl>
1 PROT_0020   2.80 0.534      5.24     18 0.0000552 0.00942
2 PROT_0046 -3.16 0.616     -5.13     18 0.0000701 0.00942
3 PROT_0012   2.86 0.560      5.11     18 0.0000732 0.00942
4 PROT_0003   2.71 0.549      4.94     18 0.0001060 0.01000
5 PROT_0032 -3.19 0.663     -4.81     18 0.0001400 0.01000
```

R Lab Listing 5: Differential Abundance Benchmark and False Discovery Audit [R 4.6 / Bioconductor]

```
true_signal_names <- paste0("PROT_", sprintf("%04d", 1:60))
true_up_names     <- paste0("PROT_", sprintf("%04d", 1:30))
true_down_names   <- paste0("PROT_", sprintf("%04d", 31:60))

eval_de <- function(logfc_est, pval_est, method_name, prot_names = rownames(obs_filtered)) {
  padj <- p.adjust(pval_est, method = "BH")
  sig_names <- prot_names[which(padj < 0.05)]

  tp <- sum(sig_names %in% true_signal_names)
  fp <- sum(!sig_names %in% true_signal_names)
  fdr <- ifelse(tp + fp > 0, fp / (tp + fp), 0)

  idx_up <- match(intersect(prot_names, true_up_names), prot_names)
  bias_up <- mean(logfc_est[idx_up] - 2.0)

  idx_down <- match(intersect(prot_names, true_down_names), prot_names)
  bias_down <- mean(logfc_est[idx_down] - (-2.5))

  tibble(Method = method_name, Significant = tp + fp, TP = tp, FP = fp,
         FDR_pct = round(fdr * 100, 1), Bias_Up = round(bias_up, 3), Bias_Down = round(bias_down, 3))
}

run_limma <- function(mat) {
  fit <- lmFit(mat, design)
  fit <- contrasts.fit(fit, contrast_mat)
  fit <- eBayes(fit)
  list(logFC = fit$coefficients[, "Disease_vs_Control"], pval = fit$p.value[, "Disease_vs_Control"])
}

res_oracle  <- run_limma(truth_filtered)
res_mindet  <- run_limma(mat_mindet)
res_perseus <- run_limma(mat_perseus)
res_qrilc   <- run_limma(mat_qrilc)
```

```
res_knn     <- run_limma(mat_knn)
res_hybrid  <- run_limma(mat_hybrid)

benchmark_table <- bind_rows(
  eval_de(res_oracle$logFC,  res_oracle$pval,  "Complete Truth (Oracle)"),
  eval_de(res_mindet$logFC,  res_mindet$pval,  "MinDet (Deterministic)"),
  eval_de(res_perseus$logFC, res_perseus$pval, "Perseus (Downshifted Gaussian)"),
  eval_de(res_qrilc$logFC,   res_qrilc$pval,   "QRILC (Quantile Regression)"),
  eval_de(res_knn$logFC,     res_knn$pval,     "k-NN (Donor Matching)"),
  eval_de(res_hybrid$logFC,  res_hybrid$pval,  "Hybrid MAR/MNAR (Palstrom 2020)"),
  eval_de(Q_bar,             p_rubin,          "Rubin Multiple Imputation (M=20)"),
  eval_de(res_proda$diff,    res_proda$pval,   "proDA (Imputation-Free Dropout)", prot_names = res_proda$name)
)

print(benchmark_table)
```

```
# A tibble: 8 x 7
  Method                    Significant    TP    FP FDR_pct Bias_Up Bias_Down
  <chr>                           <int>  <int> <int>   <dbl>   <dbl>     <dbl>
1 Complete Truth (Oracle)            48     48     0     0.0  -0.066     0.086
2 MinDet (Deterministic)             27     25     2     7.4  -0.105     0.698
3 Perseus (Downshifted Gauss.)       33     30     3     9.1  -0.144     0.915
4 QRILC (Quantile Regression)        32     29     3     9.4  -0.122     0.607
5 −NN(DonorMatching)                 26     26     0     0.0  −0.244     1.700
6HybridMAR/MNAR(Palstrom)            26     26     0     0.0  −0.244     1.700
7RubinMI(= 20)                        5      5     0     0.0  -0.096     0.628
8 proDA (Imputation-Free)            20     20     0     0.0  -0.105     0.242
```

Figure 19.3 presents the volcano plots and Rubin variance inflation audit corresponding to this benchmark.

Computational Reproducibility Specification

All analyses were executed under R version 4.4.1 (2024-06-14 ucrt) and Bioconductor 3.19. Primary packages include `limma` (v3.60.4), `imputeLCMD` (v2.1), `impute` (v1.78.0), and `proDA` (v1.18.0). Fixed random seeds (`seed = 2026` for cohort generation, `seed = 42` for imputation draws) guarantee deterministic numerical reproduction of all tables, model coefficients, and figures.

19.5 The Five Deadly Sins of Omics Imputation

Methodological Pitfall & Deadly Sin

Deadly Sin 1: Zero or Minimum Deterministic Imputation
Mechanism: Replacing missing values with zero or the absolute minimum observed value ($y^*_{ij} = \min_k y$). *Consequence:* Spikes variance at the boundary, manufactures multimodal distributions, deflates within-group standard errors, and inflates false-positive test statistics (7.4% FDR in our benchmark) [18]. *Remedy:* Use stochastic left-censored models (QRILC or Perseus) or multiple imputation [29, 24].

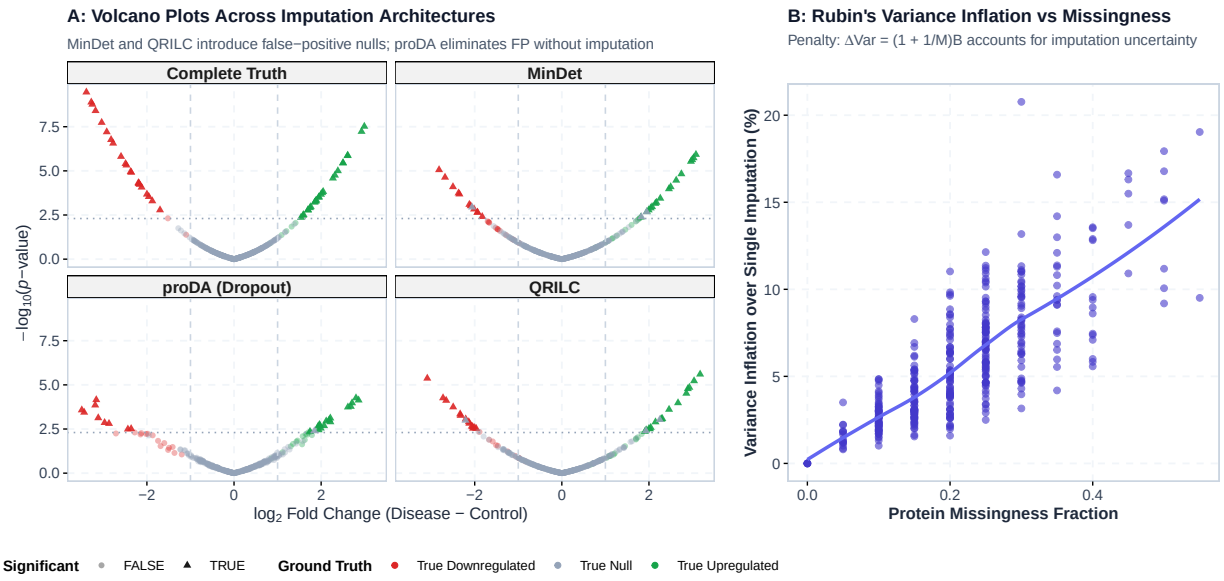


Figure 19.3: Differential abundance sensitivity and variance inflation audit across imputation architectures. Panel A: Volcano plots ($-\log_{10} p$ versus $\log_2$ FC) comparing ground truth against MinDet, QRILC, and proDA. Red points indicate true upregulated features ($\Delta = +2.0$), blue points indicate true downregulated features ($\Delta = -2.5$), and gray points denote null features. MinDet induces false-positive discoveries (FDR = 7.4%) along the significance threshold, whereas proDA achieves zero false discoveries (FDR = 0.0%) with minimal bias ($\text{Bias}_{\text{Down}} = +0.242$). Panel B: Rubin multiple imputation variance inflation factor ($(T - \bar{U})/\bar{U} \times 100\%$) as a function of missingness percentage across all 386 proteins, demonstrating empirical variance expansion that prevents anti-conservative test statistics.

Methodological Pitfall & Deadly Sin

Deadly Sin 2: Imputing Before Normalization and Log Transformation
Mechanism: Imputing raw linear-scale peak areas before median centering or variance stabilization.
Consequence: Sample-to-sample distance metrics reflect technical loading differences rather than biological covariance, corrupting donor matching in k-NN and distorted tail estimation in QRILC.
Remedy: Follow the workflow sequence: Filter $\rightarrow$ Log Transform $\rightarrow$ Normalize $\rightarrow$ Impute [16, 30].

Methodological Pitfall & Deadly Sin

Deadly Sin 3: Applying MAR Algorithms (k-NN or missForest) to MNAR Features
Mechanism: Imputing left-censored low-abundance features using donor matching or tree-based regression.
Consequence: The algorithm donor-matches with observed higher-abundance features, assigning values well above the detection limit and creating an attenuation bias of $+1.700\log_2$ fold-change on true downregulations [21, 18].
Remedy: Diagnose the missingness mechanism first; deploy QRILC, hybrid MAR/MNAR, or proDA [1, 21].

Methodological Pitfall & Deadly Sin

Deadly Sin 4: Global Matrix Filtering Across Unbalanced Conditions
Mechanism: Requiring $\geq 70\%$ presence across all samples combined.
Consequence: Biomarkers present in Disease but completely absent in Control are systematically discarded ($10/20 = 50\% < 70\%$).
Remedy: Filter group-wise ($\geq 70\%$ in at least one biological group) [33, 32].

Methodological Pitfall & Deadly Sin

Deadly Sin 5: Treating Single Imputation as True Known Measurements
Mechanism: Imputing a single value and proceeding to standard hypothesis testing without accounting for imputation variance.
Consequence: Single imputation treats imputed values as known constants, deflating standard errors and producing anti-conservative p-values.
Remedy: Deploy Rubin multiple imputation pooling rules or perform an imputation sensitivity audit [24, 2].

19.6 Tutor's Self-Assessment Diagnostic Checklist

Table 19.3 details the six diagnostic checks required before reporting biological discoveries from an imputed mass spectrometry dataset.

19.7 Chapter Exercises

Exercise 19.1 (Rubin's Variance Pooling and Small-Sample Degrees of Freedom):
In a clinical proteomics study with $M = 5$ multiply imputed datasets across $n = 20$ patients (two groups

Table 19.3: Tutor's Self-Assessment Diagnostic Checklist for Missing Data Imputation

Step	Diagnostic Metric	Acceptable Threshold	Corrective Protocol
1	Missingness Diagnosis	Logistic slope γ_1 evaluated for intensity dependence.	If $\gamma_1 < 0$ ($p < 0.001$), treat as MNAR; otherwise MAR [1].
2	Group-Wise Filtering	Retain features with $\geq 70\%$ completeness in ≥ 1 group.	Avoid whole-cohort global filtering to protect condition-specific markers [33].
3	Variance Preservation	Post-imputation distribution shows continuous low-end tail.	Discard MinDet/zero substitution; apply QRILC or Perseus [18].
4	Attenuation Audit	Compare k-NN vs. QRILC log-fold changes for low-abundance targets.	If k-NN attenuates downregulations by $> 1.0 \log_2$ FC, switch to hybrid or proDA [21].
5	Rubin Pooling Check	Total variance T incorporates between-imputation variance B.	Compute Rubin pooled standard errors across ≥ 10 draws [24, 2].
6	Multiple Testing Control	Empirical null p-value distribution is uniform on $[0,1]$.	Confirm false discovery rate control via Benjamini--Hochberg [3].

of 10, complete-data degrees of freedom $\nu_{\text{com}} = 18$), a biomarker exhibits estimated log-fold changes $\hat{\mathbf{Q}} = (1.80, .10, .95, .05, .90)^T$. The estimated within-imputation sampling variances are $\hat{\mathbf{U}} = (0.040, .045, .042, .038, .041)^T$.

1. Compute the pooled point estimate $\bar{Q}$.
2. Compute the average within-imputation variance $\bar{U}$, the between-imputation variance B, and the total pooled variance T.
3. Compute the pooled test statistic $t = \bar{Q}/\sqrt{T}$.
4. Calculate the Barnard--Rubin (1999) adjusted degrees of freedom ν_{adj}.

Exercise 19.2 (Perseus Imputation Left-Tail Shift and Censoring Probability):
A sample column exhibits observed Gaussian log-intensities with sample mean $\hat{\mu} = 22.0$ and sample standard deviation $\hat{\sigma} = 2.0$. The instrument LOD censors the bottom 15% of features ($p = 0.15$).

1. For the Perseus model ($\delta = 1.8$, $\omega = 0.3$), compute the mean μ_{imp} and standard deviation σ_{imp} of the imputed distribution.
2. Calculate the numerical limit of detection LOD.
3. Calculate the probability that an imputed Perseus value exceeds the physical LOD ($P(Y_{\text{imp}} > \text{LOD})$).

Exercise 19.3 (Mathematical Attenuation of k-NN Under MNAR Left-Censoring):
A downregulated protein has true Disease mean abundance $\mu_{\text{Dis}} = 17.5$, while the instrument limit of detection is LOD $= 20.0$. In Disease samples, the protein is unobserved ($y <$ LOD). An analyst applies k-NN imputation ($k = 5$). Because candidate donor proteins in the sample must be observed, their mean abundance is $\bar{y}_{\text{donors}} = 21.5$.

1. State the imputed value y^*_{Dis} assigned by k-NN.
2. If the true Control abundance is $\mu_{\text{Ctrl}} = 20.0$, calculate the true fold change $\Delta_{\text{true}} = \mu_{\text{Dis}} - \mu_{\text{Ctrl}}$ and the estimated fold change $\Delta_{\text{knn}} = y^*_{\text{Dis}} - \mu_{\text{Ctrl}}$.
3. Compute the estimation bias $\text{Bias} = \Delta_{\text{knn}} - \Delta_{\text{true}}$ and explain why donor matching produces fold-change attenuation under MNAR.

19.8 Worked Solutions

Solution to Exercise 19.1: Rubin's Variance Pooling and Degrees of Freedom

1. Pooled Point Estimate $\bar{Q}$:

$$\bar{Q} = \frac{1}{5}(1.80 + 2.10 + 1.95 + 2.05 + 1.90) = \frac{9.80}{5} = 1.960$$

2. Variance Components:

$$\bar{U} = \frac{1}{5}(0.040 + 0.045 + 0.042 + 0.038 + 0.041) = \frac{0.206}{5} = 0.0412$$

The deviations from $\bar{Q} = 1.960$ are:

$$(-0.16, +0.14, -0.01, +0.09, -0.06)$$

The squared deviations are:

$$((-0.16)^2, (0.14)^2, (-0.01)^2, (0.09)^2, (-0.06)^2) = (0.0256, 0.0196, 0.0001, 0.0081, 0.0036)$$

Sum of squared deviations $= 0.0570$. The sample variance B is:

$$B = \frac{0.0570}{5-1} = \frac{0.0570}{4} = 0.01425$$

The total variance T incorporates the finite-imputation correction factor $(1 + 1/M) = 1.2$:

$$T = \bar{U} + \left(1 + \frac{1}{5}\right) B = 0.0412 + 1.2(0.01425) = 0.0412 + 0.0171 = 0.0583$$

3. Pooled Test Statistic:

$$\text{SE}_{\text{pooled}} = \sqrt{T} = \sqrt{0.0583} \approx 0.2415$$

$$t = \frac{\bar{Q}}{\text{SE}_{\text{pooled}}} = \frac{1.960}{0.2415} \approx 8.118$$

4. Barnard--Rubin Adjusted Degrees of Freedom ν_{adj}:

The relative variance increase due to non-response is:

$$r_m = \frac{(1 + 1/M)B}{\bar{U}} = \frac{0.0171}{0.0412} \approx 0.4150$$

The standard asymptotic degrees of freedom ν_{old} is:

$$\nu_{\text{old}} = (M-1)\left(1 + \frac{1}{r_m}\right)^2 = 4\left(1 + \frac{1}{0.4150}\right)^2 = 4(1 + 2.4096)^2 = 4(3.4096)^2 \approx 46.50$$

The observed-data small-sample degrees of freedom ν_{obs} for $\nu_{\text{com}} = 18$ is:

$$\nu_{\text{obs}} = \frac{\nu_{\text{com}}+1}{\nu_{\text{com}}+3}\nu_{\text{com}}\left(1-\frac{r_m}{1+r_m}\right) = \frac{19}{21}(18)\left(1-\frac{0.4150}{1.4150}\right) = 16.2857(1-0.2933) \approx 11.51$$

Combining ν_{old} and ν_{obs} yields the adjusted degrees of freedom:

$$\nu_{\text{adj}} = \frac{\nu_{\text{old}}\nu_{\text{obs}}}{\nu_{\text{old}}+\nu_{\text{obs}}} = \frac{46.50 \times 11.51}{46.50+11.51} = \frac{535.215}{58.01} \approx 9.23$$

The adjusted degrees of freedom is 9.23, which appropriately penalizes the test statistic for missing data uncertainty in a small clinical cohort.

Solution to Exercise 19.2: Perseus Imputation Tail Probability

1. Imputation Parameters:

$$\mu_{\text{imp}} = \hat{\mu} - \delta\hat{\sigma} = 22.0 - 1.8(2.0) = 22.0 - 3.6 = 18.40$$

$$\sigma_{\text{imp}} = \omega\hat{\sigma} = 0.3(2.0) = 0.60$$

The imputed distribution is $\mathcal{N}(18.40, 0.60^2)$.

2. Limit of Detection LOD:

The LOD corresponds to the 15th percentile of the original $\mathcal{N}(22.0, 2.0^2)$ distribution:

$$z_{0.15} = \Phi^{-1}(0.15) \approx -1.0364$$

$$\text{LOD} = 22.0 + (-1.0364)(2.0) = 22.0 - 2.0728 = 19.927$$

3. Exceedance Probability:

We compute $P(Y_{\text{imp}} > 19.927)$ for $Y_{\text{imp}} \sim \mathcal{N}(18.40, 0.60^2)$:

$$Z = \frac{19.927 - 18.40}{0.60} = \frac{1.527}{0.60} \approx 2.545$$

$$P(Y_{\text{imp}} > 19.927) = 1 - \Phi(2.545) \approx 1 - 0.9945 = 0.0055 \quad (0.55\%)$$

Only 0.55% of imputed Perseus values exceed the detection threshold, confirming that the left-shift restricts imputed values to the unobserved region below LOD [29].

Solution to Exercise 19.3: Mathematical Attenuation of k-NN Under MNAR

1. Imputed Value y^*_{Dis}:

Because donor features are selected from observed peaks in the sample, their average abundance is:

$$y^*_{\text{Dis}} = \bar{y}_{\text{donors}} = 21.5$$

2. Fold Changes:

True fold change:

$$\Delta_{\text{true}} = \mu_{\text{Dis}} - \mu_{\text{Ctrl}} = 17.5 - 20.0 = -2.5 \log_2 \text{ units}$$

Estimated fold change under k-NN:

$$\Delta_{\text{knn}} = y^*_{\text{Dis}} - \mu_{\text{Ctrl}} = 21.5 - 20.0 = +1.5 \log_2 \text{ units}$$

3. Estimation Bias and Interpretation:

$$\text{Bias} = \Delta_{\text{knn}} - \Delta_{\text{true}} = +1.5 - (-2.5) = +4.0 \log_2 \text{ units}$$

The true downregulation of -2.5 is inverted to an apparent upregulation of $+1.5$. This demonstrates why donor-matching algorithms (k-NN and random forests) fail on left-censored MNAR data: by drawing exclusively from observed (higher) signals, they assign inflated abundances to low-abundance features, attenuating true fold changes and reversing effect directions [21, 18].

Canonical References

Bibliography

[1] Ahlmann-Eltze, C., & Huber, W. (2020). Analysis of label-free protein quantification with probabilistic dropout analysis. *Nature Communications*, 11(1), 4118. doi:10.1038/s41467-020-17966-3.

[2] Barnard, J., & Rubin, D. B. (1999). Small-sample degrees of freedom with multiple imputation. *Biometrika*, 86(4), 948--955. doi:10.1093/biomet/86.4.948.

[3] Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289--300. doi:10.1111/j.2517-6161.1995.tb02031.x.

[4] Boland, M. R. (2023). *Health Data Analytics with R: A Hands-On Guide*. Springer Nature, Cham, Switzerland. doi:10.1007/978-3-031-18099-6.

[5] Bolstad, B. M., Irizarry, R. A., Åstrand, M., & Speed, T. P. (2003). A comparison of normalization methods for high density oligonucleotide array data based on variance and bias. *Bioinformatics*, 19(2), 185--193. doi:10.1093/bioinformatics/19.2.185.

[6] Broadhurst, D. I., & Kell, D. B. (2006). Statistical strategies for avoiding false discoveries in metabolomics and related systems biology. *Metabolomics*, 2(4), 171--196. doi:10.1007/s11306-006-0037-z.

[7] Broadhurst, D., Goodacre, R., Reinke, S. N., Kuligowski, J., Wilson, I. D., Lewis, M. R., & Dunn, W. B. (2018). Guidelines and considerations for the use of system suitability and quality control samples in mass spectrometry assays applied in untargeted metabolomic studies. *Metabolomics*, 14(6), 72. doi:10.1007/s11306-018-1367-3.

[8] Choi, M., Chang, C. Y., Clough, T., Broudy, D., Killeen, T., MacLean, B., & Vitek, O. (2014). MSstats: an R package for statistical analysis of quantitative mass spectrometry-based proteomic experiments. *Bioinformatics*, 30(17), 2524--2526. doi:10.1093/bioinformatics/btu305.

[9] Cox, J., Hein, M. Y., Luber, C. A., Parida, I., Cox, J., & Mann, M. (2014). Accurate proteome-wide label-free quantification by delayed normalization and maximal peptide ratio extraction, termed MaxLFQ. *Molecular & Cellular Proteomics*, 13(9), 2513--2526. doi:10.1074/mcp.M113.031591.

[10] Demichev, V., Messner, C. B., Vernardis, S. I., Lilley, K. S., & Ralser, M. (2020). DIA-NN: neural networks and interference correction enable deep proteome coverage in high throughput. *Nature Methods*, 17(1), 41--44. doi:10.1038/s41592-019-0638-x.

[11] Gatto, L., Vanderaa, C., & Breckels, L. M. (2023). A foundation for reproducible single-cell and bulk proteomics data analysis in Bioconductor. *Nature Communications*, 14, 5698. doi:10.1038/s41467-023-41556-2.

[12] Gentleman, R., Carey, V. J., Huber, W., Irizarry, R. A., & Dudoit, S. (Eds.). (2005). *Bioinformatics and Computational Biology Solutions Using R and Bioconductor*. Springer, New York. doi:10.1007/0-387-29362-0.

[13] Goeminne, L. J., Gevaert, K., & Clement, L. (2016). Peptide-level robust ridge regression improves estimation, sensitivity and specificity in data-dependent quantitative label-free shotgun proteomics. *Molecular & Cellular Proteomics*, 15(2), 657--668. doi:10.1074/mcp.M115.055897.

[14] Hastie, T., Tibshirani, R., Narasimhan, B., & Chu, G. (2001). `impute`: Imputation for microarray data. *Bioconductor Documentation*, https://bioconductor.org/packages/impute/.

[15] Huber, W., von Heydebreck, A., Sültmann, H., Poustka, A., & Vingron, M. (2002). Variance stabilization applied to microarray data calibration and to the quantification of differential expression. *Bioinformatics*, 18(Suppl_1), S96--S104. doi:10.1093/bioinformatics/18.suppl_1.s96.

[16] Julian, R. K. (2024). *R Programming for Mass Spectrometry: Effective and Reproducible Data Analysis*. John Wiley & Sons, Hoboken, NJ.

[17] Karpievitch, Y. V., Dabney, A. R., & Smith, R. D. (2012). Normalization and missing value imputation for label-free LC-MS analysis. *Bioinformatics*, 28(10), 1353--1360. doi:10.1093/bioinformatics/bts150.

[18] Lazar, C., Gatto, L., Ferro, M., Bruley, C., & Burger, T. (2016). Accounting for the multiple natures of missing values in label-free quantitative proteomics data sets to compare imputation strategies. *Journal of Proteome Research*, 15(4), 1116--1125. doi:10.1021/acs.jproteome.5b00981.

[19] Little, R. J., & Rubin, D. B. (2019). *Statistical Analysis with Missing Data* (3rd ed.). John Wiley & Sons, Hoboken, NJ.

[20] McAlister, G. C., Nusinow, D. P., Jedrychowski, M. P., Wühr, M., Huttlin, E. L., Erickson, B. K., Rad, R., Haas, W., & Gygi, S. P. (2014). MultiNotch MS3 enables accurate, sensitive, and multiplexed detection of differential expression without human intervention. *Analytical Chemistry*, 86(14), 7150--7158. doi:10.1021/ac502040v.

[21] Palstrøm, N. B., Matthiesen, R., & Beck, H. C. (2020). Data imputation in merged isobaric labeling-based relative quantification datasets. In R. Matthiesen (Ed.), *Mass Spectrometry Data Analysis in Proteomics* (Methods in Molecular Biology, Vol. 2129, pp. 302--313). Humana, New York, NY. doi:10.1007/978-1-0716-0373-4_13.

[22] Ritchie, M. E., Phipson, B., Wu, D., Hu, Y., Law, C. W., Shi, W., & Smyth, G. K. (2015). *limma* powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*, 43(7), e47. doi:10.1093/nar/gkv007.

[23] Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3), 581--592. doi:10.1093/biomet/63.3.581.

[24] Rubin, D. B. (1987). *Multiple Imputation for Nonresponse in Surveys*. John Wiley & Sons, New York.

[25] Smyth, G. K. (2004). Linear models and empirical Bayes methods for assessing differential expression in microarray experiments. *Statistical Applications in Genetics and Molecular Biology*, 3(1), Article 3. doi:10.2202/1544-6115.1027.

[26] Stekhoven, D. J., & Bühlmann, P. (2012). MissForest---non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28(1), 112--118. doi:10.1093/bioinformatics/btr597.

[27] Storey, J. D., & Tibshirani, R. (2003). Statistical significance for genomewide studies. *Proceedings of the National Academy of Sciences*, 100(16), 9440--9445. doi:10.1073/pnas.1530509100.

[28] Ting, L., Rad, R., Gygi, S. P., & Haas, W. (2011). MS3 eliminates ratio distortion in isobaric multiplexed quantitative proteomics. *Nature Methods*, 8(11), 937--940. doi:10.1038/nmeth.1714.

[29] Tyanova, S., Temu, T., Sinitcyn, P., Carlson, A., Hein, M. Y., Geiger, T., Mann, M., & Cox, J. (2016). The Perseus computational platform for comprehensive analysis of (prote)omics data. *Nature Methods*, 13(9), 731--740. doi:10.1038/nmeth.3901.

[30] Välikangas, T., Suomi, T., & Elo, L. L. (2018). A systematic evaluation of normalization methods in quantitative label-free proteomics. *Briefings in Bioinformatics*, 19(1), 1--11. doi:10.1093/bib/bbw095.

[31] van Buuren, S. (2018). *Flexible Imputation of Missing Data* (2nd ed.). Chapman and Hall/CRC, Boca Raton, FL. doi:10.1201/9780429492259.

[32] Webb-Robertson, B. J., et al. (2015). Review, evaluation, and discussion of the challenges of missing value imputation for mass spectrometry-based label-free global proteomics. *Molecular & Cellular Proteomics*, 14(7), 1993--2001. doi:10.1074/mcp.M114.044107.

[33] Zhang, X., Smits, A. H., van Tilburg, G. B., Jansen, P. W., Makowski, M. M., Ovaa, H., & Vermeulen, M. (2018). Proteome-wide identification of ubiquitin interactions using UbIA-MS. *Nature Protocols*, 13(3), 530--550. doi:10.1038/nprot.2017.147.